

Lab Assignment 3

1. Heart Disease Prediction Using MLP
 - Check the dataset for missing values and handle, if any.
 - Display input and output features of the dataset.
 - Encode non-numeric input attributes using Label Encoder.
 - Construct an MLP with configuration 11x128x64x32x1. Use Adam optimizer and appropriate activation functions and train the model.
 - Predict heart disease for test data and display confusion matrix, accuracy, recall, precision and F1-score.
2. Iris Prediction using MLP.
 - Check the dataset for missing values and handle, if any.
 - Display input and output features of the dataset.
 - Encode output attribute using one hot encoder.
 - Shuffle the dataset and then count and display number of tuples in each class.
 - Normalize input attributes using standard scalar.
 - Split dataset into training/validation/test sets in 70:15:15 ratio.
 - Construct an MLP with configuration 4x32x16x8x3. Use Adam optimizer and appropriate activation functions and train the model.
 - Predict species of Iris flower for test data and display confusion matrix, weighted avg. accuracy, macro & micro recall, macro & micro precision and macro and micro F1-score.
3. Housing Price Prediction
 - Check the dataset for missing values and handle, if any.
 - Display input and output features of the dataset.
 - Encode non-numeric input attributes using label encoder.
 - Normalize input and output attributes using standard scalar.
 - Split dataset into training/validation/test sets in 70:15:15 ratio.
 - Construct an MLP with configuration 12x128x64x32x16x1. Use Adam optimizer and appropriate activation functions and train the model.
 - Predict house price for test data.
 - Perform inverse transformation of predicted and actual house price.
 - Compute and display RMSE, MAE and MAPE.